

Arvinth Balaji

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EDUCATION University of California Santa

Cruz, Santa Cruz, CA

Bachelor of Science (B.S.) in Electrical and Computer Engineering

September 2024 – Present
Expected Graduation May 2028

GPA: 3.78/4.0, **Dean's List:** 4 Quarters

Relevant Coursework: Physics (Thermo, Mechanics, Waves, E&M), Chemistry, Computer systems, algorithms, data structures, Circuit theormes/analysis, design, power transfer, analog filters

SKILLS & CERTIFICATIONS

Scientific & Research Skills: ModularProsthetic Systems Research (Mechanical Design & User Adaptability Studies, Data Organization&PerformanceMetrics,Technical Writing & Academic Research (LaTeX, Documentation)

Technical Skills: MachineLearningFundamentals & Artificial Intelligence (AI), Python, C++, Arduino, CAD Design (Onshape, SolidWorks),DataAnalysis(GoogleSheets, Cost & Material Feasibility Modeling, Version Control (Git/GitHub)

Certifications: Certificate of Completion – Modular Prosthetics Research, Onshape Certified Associate (CAD)

RESEARCH

Modular Prosthetics Research Paper

Aug 2023 - Jan 2024

Researcher | Under the mentorship ofnMaria Fernanda-Lugo Bolaños

- Conducted comprehensive research on modular prosthetic systems, evaluating their functionality, mechanical design, adaptability, and long-term viability while analyzing both technological and socio-economic implications in clinical and low-resource environments. Organized and interpreted performance metrics, cost models, and material feasibility data using Google
 - Sheets to deliver evidence-based insights. Produced a professional, publication-quality research paper in LaTeX featuring technical diagrams, data tables, and citations, while engaging in iterative drafting and academic review cycles with mentor Maria Fernanda Lugo Bolaños to ensure clarity, rigor, and scholarly impact.
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EXPERIENCE

OurEnergy CAD Intern

June 2026 - Present

Team Intern

- Interned at a technical consulting, engineering, and project development company focusing on renewable energy
- Drafted single-line diagrams in AutoCAD for 12 microgrid installations, mapping PV array, battery inverter, transfer switch, and utility service connections into a single reviewed drawing set.
- Sized conductors, conduit, and overcurrent protection for microgrid feeders and combiner runs, verifying every branch stayed under the 3% voltage drop target and met NEC 690 and 705 requirements.
- Developed grounding and bonding plans for battery enclosures and PV racking, coordinating equipment ground conductor sizing with the mechanical enclosure layout.

Formula Slug Aero Team

April 2025 – Present

Team Member

- Designed and developed aerodynamic components for UCSC Formula Slug's collegiate race car, contributing to aero package concepts focused on improving downforce, airflow management, and overall vehicle performance.
- Collaborated with cross-functional subteams to support the implementation, testing, and integration of aerodynamic designs, helping translate engineering ideas into manufacturable parts for competition use.
- Contributed to the design, fabrication, and on-car integration of aerodynamic elements for a Formula SAE vehicle, gaining hands-on experience in iterative engineering design, prototyping, and performance-driven implementation.

Arduino Seedling drone

Jun 2024 - Aug 2024

Researcher, Designer, and Builder

- Designed, CAD-modeled, and built a custom Arduino-powered drone equipped with a multi-sensor camera system capable of detecting color, shape, and thermal signatures.
 - Integrated computer vision algorithms to analyze real-time aerial imagery for environmental assessment and crop planning.
 - Developed a scanning and mapping system that identified viable planting zones based on vegetation density, soil temperature, and surface characteristics.
 - Enabled precision agriculture by improving seed distribution strategies and reducing resource waste, enhancing overall field efficiency.
 - Utilized tools such as Onshape for mechanical design, Arduino IDE for firmware development, and Python for image - processing
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